

KRISH BHAGIRATH

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Education

McMaster University

Hamilton, ON

Bachelor of Engineering in Computer Engineering

Sep. 2024 - 2028

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Introduction to Programming (A), Statistics for Engineering (A+), Logic Design (A)
- Awarded \$20,000+ in merit-based entrance scholarships for academic excellence.

Experience

Software Engineering Intern

May 2025 – Aug. 2025

Novologic

Toronto, ON

- Refactored **15+** core Laravel controllers, implementing the **repository pattern** for scalable architecture.
- Designed HTML/CSS email templates for CRM campaigns that generated **10+ qualified leads** via **A/B testing**.
- Optimized SQL and **Laravel Eloquent** queries on **MySQL** to improve schema organization and query efficiency.
- Built an **LLM-based** chatbot with **React** to enhance app navigation and reduce reliance on manual support.
- Increased feature test coverage from **~65% to ~95%** through systematic edge-case testing across **agile sprints**.

Software Developer

Sep. 2024 – Present

McMaster Medical Engineering Design Team (Med-T)

Hamilton, ON

- Designed **CradleWatch**, an embedded infant distress-detection system on **Raspberry Pi**, enabling continuous real-time monitoring via a camera-based **Flask** web dashboard and automated caregiver alerts using Twilio SMS.
- Built an NLP-based **CT-scan decision model** using ClinicalBERT to identify unnecessary diagnostic imaging.

Machine Learning Developer

Oct. 2025 – Present

McMaster Aerial Robotics & Drone Club

Hamilton, ON

- Designing **real-time** circle detection algorithms with **Python & YOLO** to perform mission-specific vision tasks.
- Developing software-in-the-loop (**SITL**) drone simulations on **Linux VM** with **DroneKit** to validate autonomy logic.
- Building a **2D LiDAR** wall-following system on **Jetson Nano** as a lightweight alternative to SLAM-based navigation.

Projects

Hiready ↗ | *Next.js, React, C++, Presage SDK, OpenCV, Gemini API, Google Speech-to-Text, ElevenLabs*

- Engineered a **feedback-driven** interview platform using **Next.js and React** for voice-driven mock interviews.
- Integrated **Google Speech-to-Text** and **ElevenLabs** voice agent for live transcription and conversational audio.
- Developed backend signal-processing logic with the **C++ Presage SDK** for webcam-derived physiological stress.
- Designed **REST/JSON-based** service communication to route speech content and physiological stress signals into the **Gemini API** for personalized interview feedback and stress-management guidance.

DeepFakeDetector ↗ | *Python, PyTorch, CNN, Vision Transformer, FFT, Next.js, FastAPI*

- Developed a **4-branch** hybrid deepfake detection model combining CNN, ViT, FFT, and gradient-based features for **cross-generator generalization**, achieving up to **96.7%** accuracy on AI-GenBench and OpenFake benchmarks.
- Enhanced ViT performance by **14%** via fine-tuning DeiT models with MLP heads and differential learning rates.
- Implemented a **Next.js and FastAPI** web app, utilizing **Grad-CAM** for visual heatmaps of deepfake predictions.

CourtML ↗ | *Python, React, scikit-learn, GitHub Actions, Pandas, NumPy*

- Developed an ML pipeline using HistGradientBoostingClassifier on **8 years** of NBA data to predict game outcomes.
- Improved model performance by **20%** through iterative experimentation with **5** regression and tree-based models.
- Reduced **200+** features to **50 predictors** using 10-game rolling windows and gradient-boosting feature importance.
- Deployed and automated an **ETL-driven** prediction pipeline using **GitHub Actions CI/CD** and **NBA API** data.
- Built a **React.js website** displaying prediction confidence, model evaluation, and decision statistics with **Chart.js**.

Technical Skills

Programming Languages: Python, Java, C, C++, JavaScript, PHP, SQL, Verilog, Linux (Ubuntu)

Frameworks & Libraries: React, Laravel, Scikit-learn, Numpy, Pandas, PyTorch, OpenCV

Other: Git, Visual Studio Code, Postman, Arduino, Raspberry Pi, FPGA, AI/ML, Cloud Infrastructure